PROBLEMA 1

procesor pe 8 biti
dim cache 8 bytes
dim linie 2 bytes

$T_{hit} = 10$ ms
$T_{miss} = 100$ ms

```
        ldi R19, 0x08
loop:   lds R20, 0x01
        lds R21, 0x03
        lds R22, 0x05
        lds R23, 0x07
        lds R24, 0x09
        dec R19
        cpi R19, 0
        brne loop
```

de 7 ori

a) cache mapat direct

	0	1
0	0x00 ~~0x08~~ ~~0x00~~ 0x08	0x01 ~~0x09~~ ~~0x01~~ 0x09
1	0x02	0x03
2	0x04	0x05
3	0x06	0x07

lds R20, 0x01	0000 0001	Miss	Miss
lds R21, 0x03	0000 0011	Miss	Hit
lds R22, 0x05	0000 0101	Miss	Hit
lds R23, 0x07	0000 0111	Miss	Hit
lds R24, 0x09	0000 1001	Miss	Miss

→ am 1 bit de coloana (pt că am 2 coloane)
→ am 2 biți de ~~eticheta~~ (pt că am 4 linii)

$$t_{final} = 5 \cdot T_{miss} + 7(2 \cdot T_{miss} + 3 \cdot T_{HIT})$$
$$= 500 + 7 \cdot (200 + 30) = 2110 \text{ ms}$$

b) cache mapat set asociativ cu 2 căi și LRU

		0	1
0	0	~~0x00~~ ~~0x08~~ ~~0x04~~ 0x04	0x01 ~~0x09~~ ~~0x05~~ 0x05
	1	~~0x04~~ ~~0x00~~ 0x08	~~0x05~~ 0x01 0x09
1	0	0x02 0x02	0x03 0x03
	1	0x06 0x06	0x07 0x07

lds R20, 0x01	0000 0001	M	M	M
lds R21, 0x03	0000 0011	M	H	H
lds R22, 0x05	0000 0101	M	M	
lds R23, 0x07	0000 0111	M	H	
lds R24, 0x09	0000 1001	M	M	

$$T_{final} = 5 \cdot T_{miss} + 7(3 \cdot M + 2 \cdot H)$$
$$= 500 + 7(300 + 20) = 500 + 7 \cdot 320$$
$$= 500 + 2240 = 2740 \text{ ms}$$

c) cache complet associatif en FIFO

0	1
0x00 0x07 0x02	0x01 0x09 0x03
0x02 0x04	0x03 0x05
0x04 0x08	0x05 0x07
0x06 0x01	0x07 0x09

lds R_{20}	0x01	0000 0001	1	M
lds R_{21}	0x03	0000 0011	1	M
lds R_{22}	0x05	0000 0101	1	M
lds R_{23}	0x07	0000 0111	1	M
lds R_{24}	0x09	0000 1001	1	M

$T_{final} = 8 \cdot 5 \cdot T_{ms} = 4000\ ms$

PROBLEMA 2

procesor pe 8 biti,
dim cache 8 bytes
dim linie 8 bytes

$t_{hit} = 40$ ns

$t_{miss} = 160$ ns

write through + no allocate

RAM

RAM[0x00] = 0	
RAM[0x01] = ~~1~~	0
RAM[0x02] = ~~2~~	4
RAM[0x03] = ~~3~~	16
RAM[0x04] = 4	
RAM[0x10] = 16	
RAM[0x11] = ~~17~~	18
RAM[0x12] = 18	
RAM[0x13] = ~~19~~	18
RAM[0x14] = 20	

load R19, 0x04 = 4
load R20, 0x00 = 0
load R21, 0x10 = 16
store 0x02, R19
store 0x03, R21
load R19, 0x01 = 1
store 0x01, R20
load R19, 0x12 = 18
store 0x13, R19
store 0x11, R19

a) mapat direct

	0	1
0	~~0x00~~ ~~0x10~~ 0x00	~~0x01~~ ~~0x11~~ 0x01
1	0x12	0x13
2	0x04	0x05
3		

load R19 0x04 0000 0|1|0 Tmiss
load R20 0x00 0000 0|0|0 Tmiss
load R21 0x10 0001 0|0|0 Tmiss
store 0x02, R19 → RAM[0x02] = 4
store 0x03, R21 → RAM[0x03] = 16
load R19, 0x01 0000 0|0|1 tmiss
store 0x01, R20 → RAM[0x01] = 0
load R19, 0x12 → 0001 0|0|1|0 tmiss
store 0x13, R19 → RAM[0x13] = 18
RAM[0x11] = 18

b) set asoc cu 2 căi cu FIFO

RAM[0x00] = 0
RAM[0x01] = ~~10~~
RAM[0x02] = ~~2~~4
RAM[0x03] = ~~3~~16
RAM[0x04] = 4
RAM[0x10] = 16
RAM[0x11] = ~~17~~18
RAM[0x12] = 18
RAM[0x13] = ~~19~~18
RAM[0x14] = 20

	0	1
0	0x04 0x10	~~0x05~~ 0x11
	0x00	0x01
1	0x12	0x13

load R_{19} 0x04 0000 0110 T_m
load R_{20} 0x00 0000 0000 T_m
load R_{21} 0x10 0001 0000 T_m
store 0x02 R_{19} → RAM[0x02] = 4
store 0x03 R_{21} → RAM[0x03] = 16
load R_{19} 0x01 0000 0001 T_H
store 0x01 R_{20} → RAM[0x01] = 0
load R_{15} 0x12 0001 0010 T_m
store 0x13 R_{19} → RAM[0x13] = 18
store 0x11 R_{19} → RAM[0x11] = 18

$T_{final} = 4M + H = 4 \cdot 160 + 40 = 640 + 40 = 680ns$

c) complet asociativ cu LRU

RAM[0x00] = 0
RAM[0x01] = ~~1~~0
RAM[0x02] = ~~2~~4
RAM[0x03] = ~~3~~16
RAM[0x04] = 4
RAM[0x10] = 16
RAM[0x11] = ~~17~~18
RAM[0x12] = 18
RAM[0x13] = ~~19~~18
RAM[0x14] = 20

0	1
0x04	0x05
0x00	0x01
0x10	0x11
0x12	0x13

load R_{19} 0x04 0000 0010 T_{miss}
load R_{20} 0x00 0000 0000 T_{miss}
load R_{21} 0x10 0001 0000 T_{miss}
store 0x02 R_{19} → RAM[0x02] = 4
store 0x03 R_{21} → RAM[0x03] = 16
load R_{19} 0x01 0000 0001 → T_{Hit}
store 0x01 R_{20} → RAM[0x01] = 0
load R_{19} 0x12 0001 0010 → T_{Miss}
store 0x13 R_{19} → RAM[0x13] = 18
stor 0x13 R_{19} → RAM[0x11] = 18

$T_{final} = 4M + H = 680 ns$

PREGATIRE CN

procesor pe 8 biti
dim cache 8 bytes
dim linie → 2 bytes

T_{hit} = 10ms

T_{miss} = 100 ms

---

a) mapat direct
b) set asoc cu 2 căi cu LRU
c) Full asociativ FIFO

a)

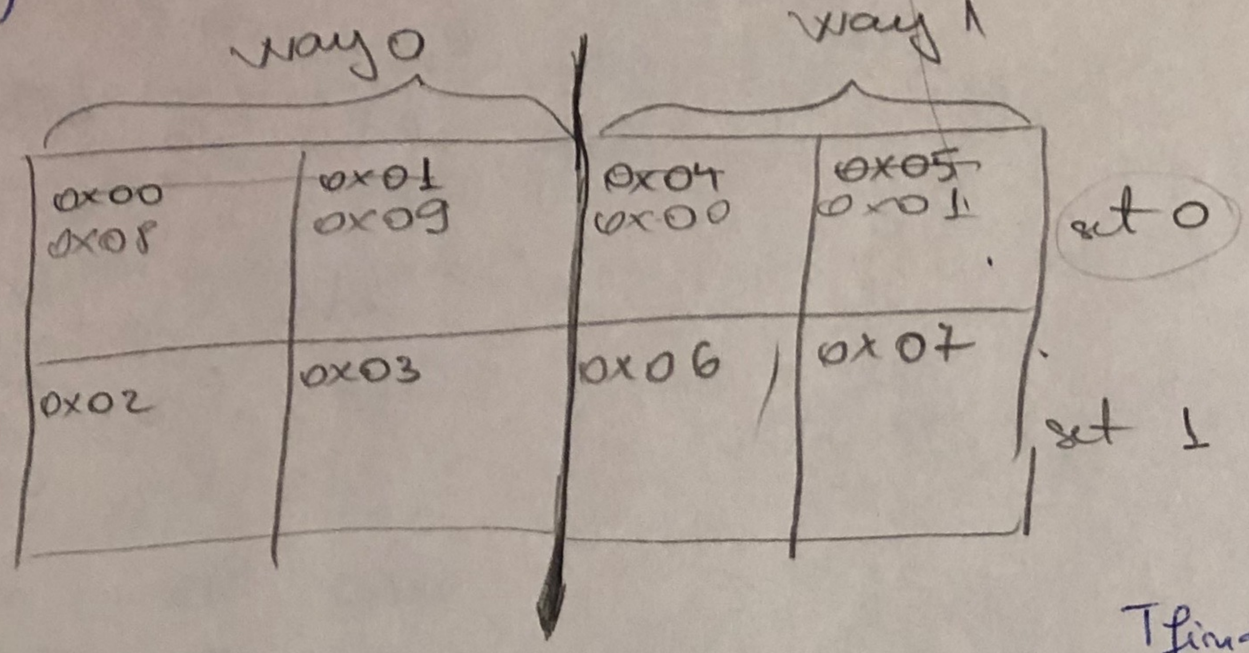
	0	1
0	~~0x00~~ 0x08	~~0x01~~ 0x09
1	0x02	0x03
2	0x04	0x05
3	0x06	0x07

```
edi R19, 0x08
loop: lds R20, 0x01
      lds R21, 0x03
      lds R22, 0x05
      lds R23, 0x07
      lds R24, 0x09
      dec R19
      cpi R19, 0
      brne loop.
```

t final = 5 · Tmiss + 7(2 · Tmiss + 3 Thit)
= 500 + 7(200 + 30)
= 500 + 7 · 230 = 500 + 1610
= 2110

lds R20, 0x01 (linie 1 poz în linie) 0000 0,00,1 (tag)	Tmiss	Tmiss
lds R21, 0x03 0000 0,01,1	Tmiss	Thit
lds R22, 0x05 0000 0,10,1	Tmiss	Thit
lds R23 0x07 0000 0,11,1	Tmiss	thit
lds R24 0x09 0000 1,00,1	Tmiss	Tmiss

b)



lds 0x01 — loc în linie — Tmiss | Tmiss
0000 0001

lds 0x03 — set — Tmiss | Thit
0000 0011

lds 0x05 — Tmiss | Tmiss
0000 0101

lds 0x07 — Tmiss | Thit
0000 0111

lds 0x09 — Tmiss | Tmiss
0000 1001

$T_{final} = 5 \cdot t_{miss} + 7(2 \cdot T_{hit} + 3 \cdot T_{miss})$

$= 500 + 7(2 \cdot 20 + 300)$

$= 500 + 7 \cdot 320 = 2740$

~~$= 500 + 2800 = 2800$~~

c)

0	1
0x00 0x08	0x01 0x09
0x02	0x03
0x04	0x05
0x06	0x01

$T_{final} = 5 \cdot 8 \cdot t_{miss} = 5 \cdot 8 \cdot 100 = 4000$

lds 0x01
0000 0001 Tmiss

lds 0x03
0000 0011 Tmiss

lds 0x05
0000 0101 Tmiss

lds 0x07
0000 0111 Tmiss

eds 0x09
0000 1001 Tmiss

2. cache 8 octeti
linii 2 octeti

$T_{hit} = 40$ ns

$T_{miss} = 160$ ns

write through fără write allocate

în ram 0x01 → i

a) mapat direct

	0	1
0	0x00	0x01
1	0x10	0x11
	0x00	0x01
2		
3	0x04	0x05
4		

load 0x04

0000 0|1|0|0

load 0x00

0000 0|0|0|0

load 0x10

0001 0|0|0|0

RAM[0x02] = 4

RAM[0x03] = 16

load R19 0x01

0000 0|0|0|1

RAM[0x01] = 0

load R19 0x12

0001 0|0|1|0

load R19, [0x04] = 4

load R20, 0x00

load R21, 0x10 =

store 0x02, R19

store 0x03, R21

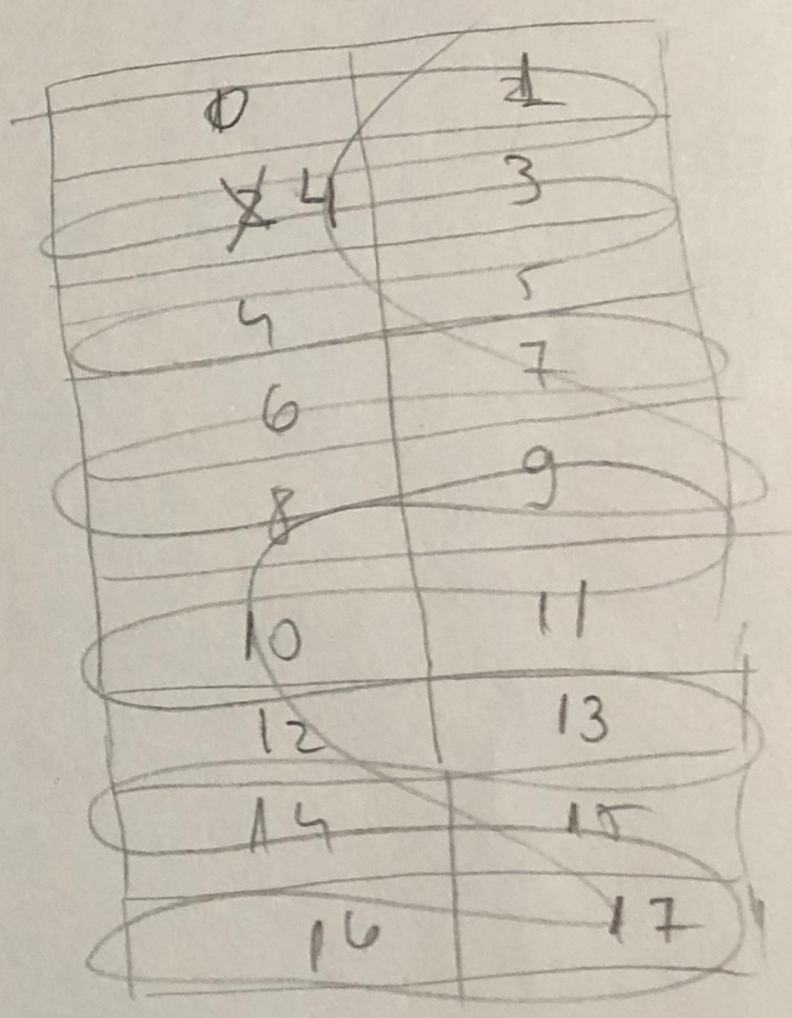
load R19, 0x01

store 0x01, R20

load R19, 0x12

store 0x13, R19

store 0x11, R19



PROBLEMA 3

procesor pe 8 biti
dim cache 8 octeti
dim linie 2 octeti
$t_{hit} = 20ns$
$t_{miss} = 230 ns$
write-back + write-allocate

a) mapare directa

RAM[0x00] = 0
RAM[0x01] = 16
RAM[0x02] = 20
RAM[0x03] = 32
RAM[0x04] = 4
RAM[0x10] = 16
RAM[0x11] = 17
RAM[0x12] = 18
RAM[0x13] = 19

	0	1
0	~~0x00/0~~ 0x10/16 ~~0x00/0~~ 16 0x10/16	~~0x01/1~~ 0x11/17 ~~0x01/1~~ 16 0x11/17 18
1	~~0x02/20~~ 0x12/18	~~0x03/32~~ x 0x13/19
2		
3		

load R_{19}, [0x00]
load R_{20}, [0x10]
load R_{21}, [0x02]
load R_{22}, [0x03]
store [0x02] R_{19}
store [0x03] R_{21}
load R_{19}, [0x01]
store [0x01], R_{20}
store [0x13], R_{22}
load R_{23} [0x12]
store [0x11], R_{23}

$R_{19} = 01$
$R_{20} = 16$
$R_{21} = 2$
$R_{22} = 3$
$R_{23} = 18$

load R_{19} 0x00 0000 0|00|0 Tmiss
load R_{20} 0x10 0001 0|00|0 Tmiss
load R_{21} 0x02 0000 0|01|0 Tmiss
load R_{22} 0x03 0000 0|01|1 Thit
store [0x02] R_{19} -> hit
store [0x03] R_{21} -> hit
load R_{19} 0x01 - Tmiss
store [0x01], R_{20} -> hit
store [0x13], R_{22} -> miss 0001 0|01|1
load R_{23} [0x12] -> hit
store [0x11], R_{23} -> miss 0001 0|00|1

5) 2 ~~way~~ set asociativ cu FIFO.

RAM[0x00] = 0
RAM[0x01] = 1
RAM[0x02] = 2
RAM[0x03] = 3
RAM[0x04] = 4
RAM[0x10] = 16
RAM[0x11] = 17
RAM[0x12] = 18
RAM[0x13] = 19

	0	1
set 0	0x00/0	0x01/~~x~~16
	0x10/16	0x11/~~17~~18
set 1	0x02/~~1~~20	0x03/~~3~~2
	0x12/18	0x13/~~19~~3

load R_{19} 0x00 0000 0000 T_M
load R_{20} 0x10 0001 0000 T_M
load R_{21} 0x02 0000 0010 T_M
load R_{22} 0x03 0000 0011 T_H
store 0x02 R_{19} - hit
store 0x03 R_{21} - hit
load R_{19} 0x01 - hit
store 0x01 R_{20} - hit
store 0x13 R_{22} 0001 0011 - miss
load R_{23} 0x12 - hit
store [0x11], R_{23} - hit

$R_{19} = 01$
$R_{20} = 16$
$R_{21} = 2$
$R_{22} = 3$
$R_{23} = 18$

$t_{final} = 4 \cdot M + 7H = 4 \cdot 230 + 7 \cdot 20 = 920 + 140 = 1060$ ms

c) complet asociativ cu LRU

RAM[0x00] = 0
RAM[0x01] = 1
RAM[0x02] = 2
RAM[0x03] = 3
RAM[0x04] = 4
RAM[0x10] = 16
RAM[0x11] = 17
RAM[0x12] = 18
RAM[0x13] = 19

0	1
0x00/0	0x01/~~x~~2
0x10/16	0x11/~~17~~18
0x02/~~1~~20	0x03/~~3~~2
0x12/18	0x13/~~19~~3

load R_{19} 0x00 0000 0000 miss
load R_{20} 0x10 0001 0000 miss
load R_{21} 0x02 0000 0010 miss
load R_{22} 0x03 0000 0011 ~~hit~~
store 0x02 R_{19} hit
store 0x03 R_{21} hit
load R_{19}, 0x01 hit
store 0x01 R_{20} hit
store 0x13 R_{22} miss 0001 0011
load R_{23} 0x12 hit
store 0x11 R_{23} hit

$R_{19} = 01$
$R_{20} = 16$
$R_{21} = 2$
$R_{22} = 3$
$R_{23} = 18$

$t_{final} = 4 \cdot miss + 7 hit = 1060$ ms

PROBLEMA 4

adrese pe 8 biti,
cache de 16 octeti
dim linie = 2 octeti,

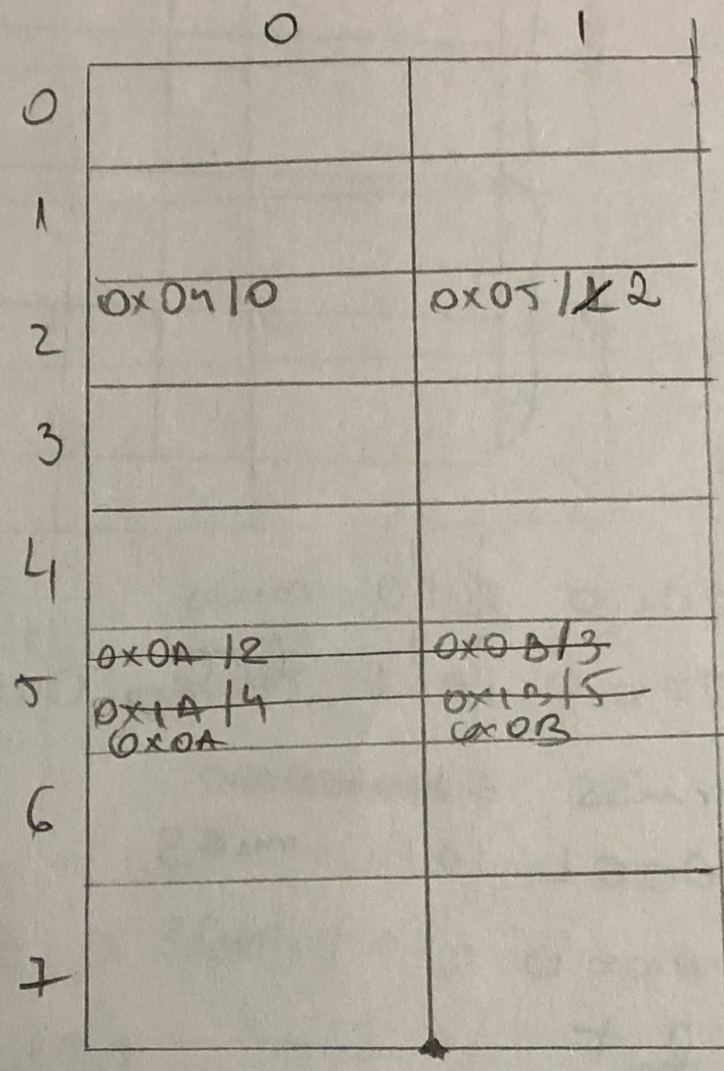
$t_{hit} = 50\,ms$

$t_{miss} = 250\,ms$

I write-through + no write allocate

a) mapare directă.

RAM[0x04] = 0
RAM[0x05] = 12
RAM[0x0A] = 2
RAM[0x0B] = 3
RAM[0x1A] = 4
RAM[0x1B] = 5
RAM[0x40] = 3
RAM[0x41] = 7



load R_{19}, 0x0A 0000 | 1010 - miss
load R_{20}, 0x0B 0000 | 1011 - hit
store 0x40, R_{20} 0100 000 | 0 - miss
load R_{20}, 0x1A 0001 10 | 10 - miss
load R_{21}, 0x05 0000 | 010 | 1 - miss
store 0x05, R_{19} - hit
load R_{19}, 0x0A 0000 | 1010 - miss

$R_{19} = 2$
$R_{20} = 34$
$R_{21} = 1$

$t_{final} = 2 \cdot h + 5\,miss = 100 + 5 \cdot 250 = 100 + 1250 = 1350\,ms$

5) set asoc cu 2 căi și LRU

RAM[0x04] = 0
RAM[0x05] = 2
RAM[0x0A] = 2
RAM[0x0B] = 3
RAM[0x1A] = 4
RAM[0x1B] = 5
RAM[0x40] = 3
RAM[0x41] = 7

	0	1
0		
1	0x0A/2	0x0B/3
	0x1A/4	0x1B/5
2	0x04/0	0x05/2
3		

$R_{19} = 2$
$R_{20} = 3$
$R_{21} =$

load R_{19}, 0x0A 0000 1|01|0 miss
load R_{20}, 0x0B 0000 1|01|1 hit
store 0x40, R_{20} - miss ~~0100 000~~
load R_{20}, 0x1A 000L 1|01|0 miss
load R_{21}, 0x05 0000 0|10|1 miss
store 0x05, R_{19} hit
load R_{19}, 0x0A hit

$t_{final} = 4 \text{ miss} + 3 \text{ hit} = 4 \cdot 250 + 3 \cdot 50 = 1000 + 150 = 1150 \text{ ns}$

c) full asoc cu FiFo

RAM [0x04] = 0
RAM [0x05] = X2
RAM [0x0A] = 2
RAM [0x0B] = 3
RAM [0x1A] = 4
RAM [0x1B] = 5
RAM [0x40] = 63
RAM [0x41] = 7

0	1
0x0A/2	0x0B/3
0x1A/4	0x1B/5
0x04/0	0x05/X2

$R_{19} = 2$
$R_{20} = 3$
$R_{21} =$

load R_{19}, 0X0A 0000 1010 - miss
load R_{20}, 0x0B - hit
store 0x40, R_{20} - miss
load R_{20}, 0x1A 0001 1010 - miss
load R_{21}, 0x05 0000 0101 - miss
store 0x05, R_{19} - hit
load R_{19}, 0x0A - hit

t_{final} = 3 hit + 4 miss = 1150 ns

PROBLEMA 5

- procesor pe 8 biți
dim cache 24 octeți
leng linie ~~3 octe~~ 4 octeți

t_{hit} = 10 ns

t_{miss} = 100 ns

write through + no write allocate

~~RA~~

a) mapare directă

RAM [0x00] = 02
RAM [0x01] = 12
RAM [0x02] = 27
RAM [0x03] = 3
RAM [0x04] = ~~4~~ 83
RAM [0x05] = 5
RAM [0x06] = 6
RAM [0x07] = ~~7~~ 75
RAM [0x08] = 8
RAM [0x09] = 9

	0	1	2	3
0	0x00/02	0x01/12	0x02/27	0x03/3
1	0x04/~~48~~3	0x05/5	0x06/6	0x07/75
2	0x08/8	0x09/89	0x0A/90	0x0B/11
3				
4				
5				

R_{19} = ~~8~~0
R_{20} = 2
R_{21} = 3
R_{22} = 5
R_{23} = 7
R_{24} = 9
R_{25} = 6

lds R_{19} 0x08 0000 1000 miss
lds R_{20} 0x02 0000 0010 miss
lds R_{21} 0x03 hit
lds R_{22}, 0x05 0000 0101 miss
sts 0x07, R_{22} hit
lds R_{24}, 0x09 hit
stg 0x04, R_{19} hit
sts 0x01, R_{20} hit
lds R_{21}, 0x03 hit
lds R_{22}, 0x05 hit
lds R_{23} 0x07 hit
sts 0x09 R_{24} hit
sts 0x02 R_{23} hit
sts 0x04, R_{21} hit
sts 0x00, R_{20}, hit.

lds R_{19}, 0x00 hit
lds R_{25}, 0x06 hit
sts 0x06, R_{19} hit

Problema 6

adresa pe 8 biti
dim cache 8 octeti
dim linie 2 octeti
T_{hit} = 40ns
t_{miss} = 160ns
write brch + write alloc~~k~~

a) mapare directă

RAM [0x00] = 0
RAM [0x01] = ~~10~~
RAM [0x02] = ~~84~~
RAM [0x03] = ~~816~~
RAM [0x04] = 4
RAM [0x10] = 16
RAM [0x11] = 17
RAM [0x12] = 18
RAM [0x13] = 19
RAM [0x14] = 20

	0	1
0	~~0x00/0~~ ~~0x10/16~~ ~~0x00/0~~ 0x10/16	~~0x01/1~~ ~~0x11/17~~ ~~0x01/1~~ ~~0x11/17~~ 18
1	~~0x02/84~~ 0x12/18	~~0x03/816~~ 0x13/~~19~~ 18
2	0x04/4	0x05/5
3		

R_{19} = ~~4~~ ~~/~~ 18
R_{20} = 0
R_{21} = 16

load R_{19} 0x04 0000 0'1'0'0 miss
load R_{20} 0x00 miss
load R_{21} 0x10 0001 0'0'0'0 miss
store 0x02 R_{19} miss 0000 0'0'1'0
store 0x03 R_{21} hit
load R_{19} 0x01 0000 0'0'0'1 miss
store 0x01, R_{20} hit
load R_{19} 0x12 0001 0'0'1'0 miss
store 0x13, R_{19} hit
store 0x11, R_{19} miss 0001 0'0'0'1

$t_{final} = 7_{miss} + 3_{hit} = 7 \cdot 160 + 3 \cdot 40 = 1120 + 120 = 1240$ ms

c) complet asoc + LRU

RAM [0x00] = 0
RAM [0x01] = 1
RAM [0x02] = 2
RAM [0x03] = 3
RAM [0x04] = 4
RAM [0x10] = 16
RAM [0x11] = 17
RAM [0x12] = 18
RAM [0x13] = 19
RAM [0x14] = 20

0	1
~~0x04/4~~ 0x12/18	~~0x05/5~~ 0x13/19/18
0x 00/0	0x01/~~1~~0
0x10/~~16~~	0x11/~~17~~18
0x02/~~2~~4	0x03/~~3~~16

$R_{19} = 4 \not\in 18$
$R_{20} = 0$
$R_{21} = 16$

load R_{19} 0x04 Miss 0000 010~~0~~
load R_{20} 0x00 Miss
load R_{21} 0x10 Miss 0001 000~~0~~
store 0x02 R_{19} Miss 0000 001~~0~~
store 0x03 R_{21} hit
load R_{19} 0x01 hit
store 0x01, R_{20} hit
load R_{19} 0x12 miss 0001 001~~0~~
store 0x13 R19 hit
store 0x11 R19 hit

b) 2 asoc cu FIFO

RAM[0x00] = 0
RAM[0x01] = 1
RAM[0x02] = 2
RAM[0x03] = 3
RAM[0x04] = 4
RAM[0x10] = 16
RAM[0x11] = 17
RAM[0x12] = 18
RAM[0x13] = 19

	0	1
0	0x00/0	0x01/1 ~~0~~
	0x10/16	0x11/~~17~~ 18
1	0x02/~~2~~4	0x03/~~3~~16
	0x12/18	0x13/~~19~~ 18

$R_{19} = 4$ ~~5~~ 18
$R_{20} = 0$
$R_{21} = 16$

load R_{19} 0x04 miss 0000 00~~1~~0
load R_{20} 0x00 miss
load R_{21} 0x10 miss 0001 0000
store 0x02 R_{19} miss 0000 0010
store 0x03 R_{21} HIT
load R_{19} 0x01 hit
store 0x01, R_{20} hit
load R_{19} 0x12 miss 0001 00~~1~~10
store 0x13, R_{19} hit
store 0x11, R_{19} hit

$t_{final} = 5_{miss} + 5_{hit} = 5 \cdot 160 + 5 \cdot 40 = 200 + 800 = 1000_{ns}$